CSE 412 Proposal - Phase 1

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Application Requirements

We plan to make “Repair Matrix”, a customer relationship management system for electronics repairs. Repair Matrix would help electronics repair shops easily manage their repair orders, statuses, and inventory. The main users would be customers, repair technicians, and the shop manager(s).

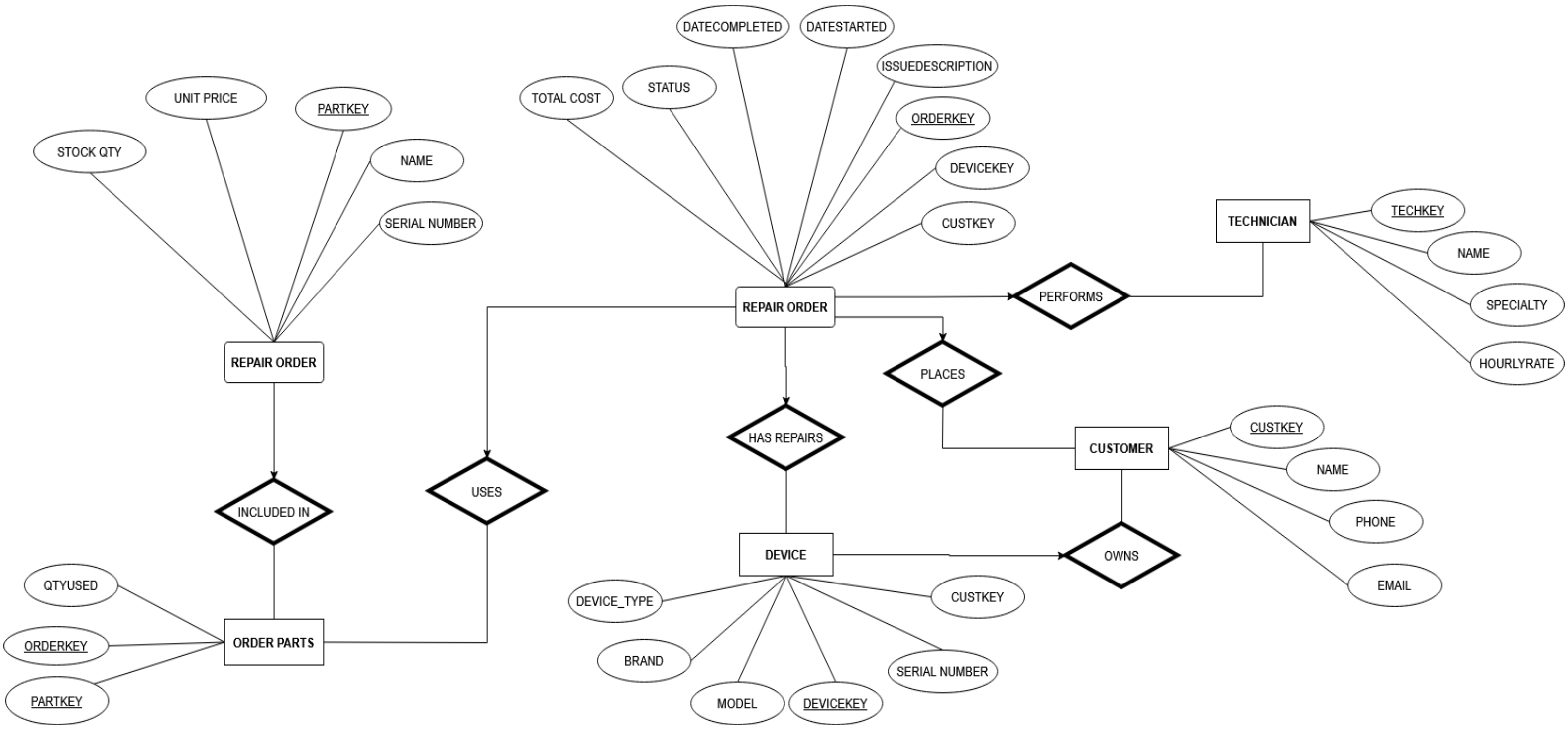
The customers can place an order to repair by specifying their name, phone number, e-mail address, and their device brand, model, type, etc. Customers can also see the status of their repair on the website. The repair technician can order parts, see the inventory of parts they possess, update repair status and complete pending orders. They receive a notification when the technician starts the repair, and another when the repair is completed.

The manager(s) can see all of this and update/fix things when necessary, for example in case the technician makes a mistake and marks an order as completed instead of pending, the manager can correct this.

When a customer places an order and enters the device details (including the device brand, model, and a description of the issue), they are quoted a repair price based on the problem (like a screen repair, motherboard issue, etc). If they choose to accept the quoted price, the status appears as “received” and is assigned an order number (ORDERKEY) and the device is assigned a key to track it (DEVICEKEY).

The technician with the least amount of orders receives the ticket for repair. If there are multiple such technicians, a random one receives the ticket. When a technician starts a repair, the status is updated to “pending”. There can only be one technician assigned to a repair, but one technician can have multiple devices to repair. The technician then orders parts if necessary, updates the shop’s inventory of parts (which each have a PARTKEY) based on what was used, and repairs the device. The order status is then updated to “completed” and the customer is charged the amount specified during the quote. Only the technician and manager are allowed to change the order status.

ER Diagram



Implementation Plans

We plan to use PostgreSQL as the database management system to store the orders and inventory. When a customer accepts the quoted repair price and places an order, the database is updated to store the customer's order as pending, and the customer's device as in the shop's possession. When a technician uses a part to repair the device or orders a part, the inventory database is modified to reflect the increase or decrease of the part quantity. When a repair is completed, the database is updated and the application receives the data (of completion) from the database.

The application is in the form of a website which we will build using Python + Flask for its operations, and the HTML/css will be rendered to Flask using Jinja. The backend (python) code sends requests to the database, which runs SQL queries on it. The results are sent back to the website and are then displayed to the users.

Link to the YouTube video

https://youtu.be/vxVkol8DFVY